

mahg_bigdata_201

SAFe and PI Planning for the ManOxCo Oxygen Reliability Program

Manuel Hernandez Giuliani

Independent Researcher and Instructor

manuelhernandezgiuliani.com

Publication Date: 2026-04-19

Publication No.: MAHG-BD-PAPER-201-20260419-01

Abstract

Large-scale data-platform initiatives require more than sound architecture; they also require a delivery model capable of coordinating teams, dependencies, risks, and measurable outcomes across a shared planning horizon. This requirement becomes more acute in industrial settings where delays, unclear ownership, or integration failures can carry operational and safety consequences. Using the ManOxCo Oxygen Reliability Transformation Program as a practical frame, this article examines the value of the *Scaled Agile Framework* (SAFe) and Program Increment (PI) Planning for complex Big Data delivery. It analyzes why scaled coordination is necessary, explains how strategic intent is decomposed into capabilities, features, enablers, and stories, and shows how estimation, capacity, dependency management, Definition of Done, and GitHub execution can be integrated into one planning model. The central argument is that SAFe becomes most useful when it is not treated as generic process overhead, but as an execution architecture that links business-critical outcomes, platform workstreams, and governance evidence within a shared delivery model (Hernandez Giuliani, 2024a, 2024b, 2026a, 2026b; Ulfes et al., 2024).

Keywords: SAFe, PI Planning, Big Data delivery, agile coordination, ManOxCo, GitHub, dependency management, value flow

1. Introduction

Modern Big Data programs are rarely single-team efforts. They involve data engineering, architecture, governance, analytics, infrastructure, security, and business stakeholders who must deliver compatible outputs under time, quality, and budget constraints. In industrial cases such as ManOxCo, this complexity is heightened because the program does not exist to

produce dashboards alone; it exists to support operational reliability, prevent hospital oxygen dry-outs, and improve planning across production, storage, delivery, and consumption (Hernandez Giuliani, 2026a, 2026b). Under these conditions, fragmented delivery becomes an architectural risk in its own right.

Traditional agile methods such as Scrum and Kanban remain valuable at team level, but they do not automatically solve cross-team synchronization, architectural dependency management, or medium-horizon commitment for enterprise-scale data work. When several teams must deliver infrastructure, ingestion, medallion pipelines, AI-enabled decision support, governance controls, and financial analysis in coordinated fashion, the risk is not merely slow execution. The deeper risk is that technically correct local work fails to compose into an end-to-end platform solution (Ulfes et al., 2024).

In this setting, the *Scaled Agile Framework* (SAFe) is useful because it provides a shared planning language for coordinating teams around a common mission, explicit dependencies, and realistic commitments. This article interprets SAFe not as an abstract management framework, but as a practical delivery model for the ManOxCo PI-1 baseline. The central concept guiding the discussion is therefore **execution architecture**: the proposition that successful data-platform delivery depends on organizing work so that planning, governance, and technical implementation reinforce each other rather than drift apart.

2. Objectives

This article pursues four principal objectives. First, it explains why the ManOxCo PI-1 program requires structured scaled coordination rather than isolated agile team activity. Second, it analyzes how SAFe concepts such as PI Planning, value hierarchy, estimation, and dependency management can be used to govern a multi-team data-platform initiative. Third, it maps these concepts to the actual ManOxCo workstreams, including infrastructure, IT data, OT streaming, medallion modeling, governance, AI decision support, and financial optimization. Fourth, it shows how GitHub can be used as an execution environment for traceable delivery commitments (Hernandez Giuliani, 2024a, 2024b, 2026a).

3. Main Discussion

3.1. Why Structured Coordination Is Required for ManOxCo

The ManOxCo PI-1 program is a useful example of why scaled coordination matters. The target solution must integrate production and storage telemetry, hospital consumption data, delivery records, truck and plant master data, governance controls, and a decision-support layer capable of reducing dry-out risk. Those goals cannot be delivered by one backlog lane alone because they involve distinct workstreams with different cadences, different validation requirements, and strong mutual dependencies. OT streaming cannot create value without platform infrastructure; predictive intelligence cannot be credible without cleaned and governed gold datasets; executive decisions cannot be defended without lineage, quality checks, and operational traceability (Hernandez Giuliani, 2026a, 2026b).

In such settings, the failure mode is often not lack of effort, but lack of synchronized visibility. Teams may optimize locally while the end-to-end program remains blocked by unresolved dependencies, unclear ownership, or mismatched assumptions about readiness. SAFe responds to this problem by creating a shared cadence, a common commitment horizon, and explicit mechanisms for dependency and risk management. In this sense, SAFe is best understood as a coordination architecture for delivery at scale rather than simply a collection of meetings (Hernandez Giuliani, 2024a; Ulfes et al., 2024).

3.2. SAFe Operating Model for the PI-1 Program

Within the PI-1 simulation, the **Agile Release Train** can be interpreted as the coordinated delivery structure that brings together the principal ManOxCo workstreams under one mission: prevent future oxygen dry-out incidents through governed data and decision capability. The ART therefore does not represent a single technical team, but a synchronized delivery model composed of platform, domain, and governance contributors.

For ManOxCo, the minimum coordinated workstreams are the following: infrastructure; IT data ingestion; OT data ingestion; medallion and dimensional modeling; governance and quality; AI decision support; and financial optimization. These workstreams should not be interpreted as isolated silos, but as bounded responsibilities that must converge during the same Program Increment. PI Planning becomes the moment where teams align on what can be delivered, what must be sequenced first, and which risks require escalation before implementation starts (Hernandez Giuliani, 2024b, 2026a).

3.3. PI Planning Foundations and Cadence

PI Planning is the principal alignment event in this model. Its function is to translate business-critical outcomes into a realistic short-to-medium planning horizon shared by all participating teams. For the ManOxCo case, that means deciding not only which features are important, but in what order they become technically and operationally viable. An OT streaming feature may appear strategically urgent, for example, but it cannot be committed without compatible infrastructure, schema rules, and a validated downstream consumption path.

The value of PI Planning lies in making such constraints explicit. Participants define objectives, review dependencies, compare desired scope with actual team capacity, and identify risks that threaten the increment. Instead of allowing architecture, governance, and engineering concerns to surface late as blockers, PI Planning forces them into the commitment conversation early. This is particularly important in Big Data programs, where platform readiness and data quality often determine whether nominal feature progress is actually usable.

3.4. Translating Strategy into Capabilities, Features, Enablers, and Stories

One of SAFe's most useful strengths is the way it decomposes strategy into manageable delivery units. In the ManOxCo case, a **capability** might be framed as oxygen reliability intelligence across production, storage, and hospital consumption. A **feature** under that capability might be a hospital dry-out risk dashboard, a near-real-time telemetry ingestion flow, or a validated gold-layer inventory product. **Enablers** then represent the architectural and technical preconditions for those features, such as message-bus setup, schema validation, identity controls, medallion pipeline scaffolding, or observability instrumentation. **User stories** finally describe the smallest negotiable units of implementation and validation.

This hierarchy matters because not all valuable work is directly user-visible. In Big Data programs, some of the most important delivery items are enabling structures that reduce future risk and make later functionality credible. If a team attempts to commit only user-facing outputs while underinvesting in infrastructure, contracts, or validation, the resulting increment often looks productive but remains operationally fragile. The hierarchy of capabilities, features, enablers, and stories therefore protects the program from mistaking visible output for deployable value (Hernandez Giuliani, 2024a, 2024b).

3.5. Estimation, Capacity, Dependencies, and ROAM Risk Management

In enterprise data work, estimation must reflect more than workload volume. It must also capture technical uncertainty, data dependency complexity, environment readiness, and validation effort. For this reason, relative estimation through story points remains useful: it discourages false precision while still enabling the team to compare scope against capacity. In the ManOxCo setting, this is important because similar-looking tasks can differ sharply in risk. A batch ingestion story and an OT streaming story may appear comparable in size, yet their operational consequences, testing requirements, and dependency surfaces are very different.

Capacity planning performs a complementary function. Historical velocity gives a practical anchor for how much a team can actually commit to within a sprint or PI. This prevents planning from becoming a wish list disconnected from execution reality. At PI level, the same logic extends across teams: if one workstream is under-resourced, downstream commitments that depend on it should be challenged or re-sequenced rather than allowed to fail later in silence.

Dependency management is equally central. The PI-1 program contains clear dependency chains: OT telemetry depends on infrastructure and messaging readiness; gold-layer analytics depend on silver validation; AI decisions depend on reliable gold products and governance constraints. These relationships should be visible during planning through dependency boards and explicit sequencing notes, not discovered ad hoc during execution.

ROAM risk management strengthens this discipline. Risks should be categorized as resolved, owned, accepted, or mitigated, with named accountability and explicit follow-up. In ManOxCo, typical ROAM candidates include delayed OT connectivity, uncertain schema quality in source data, observability gaps, incomplete governance controls, and unclear readiness for AI-assisted recommendations. Treating these risks formally keeps the PI commitment intellectually honest rather than aspirational.

3.6. Acceptance Criteria, Definition of Done, and Governance Evidence

A scaled planning model fails if it accelerates delivery while weakening quality controls. For this reason, every ManOxCo feature and story should include explicit acceptance criteria tied to data behavior, not just code completion. A telemetry ingestion story, for example, is not complete merely because a connector runs. It should satisfy criteria for data freshness, schema conformance, error handling, logging, and downstream availability.

The **Definition of Done** then supplies the universal baseline across teams. In the PI-1 context, that baseline should include code committed, validation checks implemented, README or delivery documentation updated, architectural decisions justified where relevant, and quality or governance evidence attached. This is especially important in data-platform programs because work can appear technically implemented while still being unusable due to weak lineage, missing tests, undocumented assumptions, or absent operational monitoring. The DoD therefore acts as both a technical control and a governance checkpoint.

3.7. GitHub Operational Mapping for PI-1 Delivery

SAFe becomes practical only when it is translated into daily execution tooling. Within GitHub, a useful mapping for the ManOxCo program is straightforward. An **epic** can be managed as a parent issue representing a large bounded outcome. A **feature** can be managed as a standard issue tied to a PI milestone. An **enabler** can be managed as a technically labeled issue that supports architecture, platform readiness, or governance. **User stories** can then be derived as smaller implementation issues or tracked subtasks under the feature.

Milestones support PI and sprint structure, while labels support filtering by workstream, dependency type, governance relevance, or risk status. Pull requests can provide the evidence trail for Definition of Done through links to issues, validation output, documentation updates, and reviewer approval. This mapping is important because it turns planning into traceable execution. The goal is not to mimic SAFe terminology mechanically, but to preserve alignment, evidence, and visibility from planning artifact to implementation artifact.

3.8. Expected Outputs of the PI-1 Planning Cycle

If PI Planning is working correctly, it produces more than a schedule. For the ManOxCo program, the expected outputs are a set of PI objectives, a workstream-level feature set, prioritized enablers, an explicit dependency map, a ROAM risk set, and realistic capacity-aware commitments for each participating team. The output should also make clear which architectural prerequisites must be completed before high-value case outcomes such as dry-out prediction or operational alerting can be trusted.

This final point is essential. The success of the increment should be measured by whether the program has advanced the end-to-end oxygen reliability mission, not simply by whether many tickets were closed. A delivery model is mature when it protects the integrity of the program outcome while still enabling iterative progress.

4. Conclusion

The ManOxCo PI-1 case shows why scaled delivery discipline is necessary in modern Big Data programs. The challenge is not simply to build pipelines or dashboards, but to coordinate infrastructure, ingestion, modeling, governance, AI support, and operational planning in a way that produces trustworthy and usable outcomes. Under these conditions, SAFe and PI Planning are valuable because they provide a common execution language for translating strategic intent into realistic, dependency-aware commitments.

The framework becomes especially useful when adapted to the actual program rather than taught in isolation. In the ManOxCo context, the hierarchy of capabilities, features, enablers, and stories helps preserve strategic coherence; capacity, dependency, and ROAM mechanisms keep commitments realistic; and GitHub-based traceability turns planning into inspectable delivery. The broader lesson is that scaled agile coordination should be treated as part of the architecture of execution, not as an administrative add-on to technical work.

5. Discussion Highlights

Four key findings emerge from the discussion. First, industrial Big Data programs such as ManOxCo require structured multi-team coordination because technical workstreams are highly interdependent. Second, PI Planning is useful because it makes dependencies, capacity limits, and risk visible before execution rather than after failure. Third, the hierarchy of capabilities, features, enablers, and stories protects the program from underestimating critical architectural work. Fourth, GitHub can serve as a practical execution environment for SAFe concepts when issues, milestones, evidence, and reviews are used to preserve traceability.

6. Glossary

Agile Release Train (ART): An organizational structure composed of multiple agile teams that plan and deliver solutions in a synchronized manner.

Dependency board: A planning artifact used to make cross-team sequencing and blocking relationships visible.

Definition of Done (DoD): A shared agreement establishing the minimum universal conditions required for work to be considered complete.

Enabler: Technical or architectural work required to support future functionality or reduce system constraints.

INVEST: A quality criterion for user stories: independent, negotiable, valuable, estimable, small, and testable.

PI Planning: A collaborative *Program Increment* planning event aimed at aligning objectives, dependencies, and capacity.

ROAM: A risk-management approach that classifies risks as resolved, owned, accepted, or mitigated.

Team capacity: The volume of work that a team can realistically complete in a sprint or planning horizon based on historical delivery ability.

Story points: A relative estimation unit that considers effort, complexity, and uncertainty.

7. References

Hernandez Giuliani, M. (2024a). *Big Data: Clase 2 - SAFe in action: PI planning simulation* [Course material].

Hernandez Giuliani, M. (2024b). *Big Data: Clase 7 - Big Data platforms organizations and stakeholders (politics) / Understanding SAFe* [Course material].

Hernandez Giuliani, M. (2026a). *ManOxCo platform - Program baseline repository: Oxygen Reliability Transformation - PI-1* [Program baseline]. [slides/docs/ManOxCo/manoxco_pi_1.md](#).

Hernandez Giuliani, M. (2026b). *ManOxCo (Man Oxygen Company) - Case study* [Case study]. [slides/docs/ManOxCo/manoxco_use_case_en.md](#).

Ulfes, R., et al. (2024). *Exploring the organizational models for data science in agile software development: Challenges and strategies from a multi-case study. Proceedings of the 57th Hawaii International Conference on System Sciences.*

Appendix A. Self-Assessment Exercises

1. In the ManOxCo PI-1 context, what is the primary purpose of PI Planning?

- A. To eliminate the need for architecture and governance work.
- B. To align teams, dependencies, risks, and realistic commitments around the oxygen reliability mission.
- C. To turn all work into independent tasks with no sequencing needs.
- D. To replace operational planning with executive reporting.

2. Which of the following is the best example of an enabler in the ManOxCo program?

- A. A validated streaming ingestion backbone for hospital telemetry.
- B. A final board presentation already based on complete data products.
- C. A monthly invoice sent to a customer.
- D. A hospital's oxygen consumption itself.

3. Why is dependency management especially important in the ManOxCo PI-1 program?

- A. Because each workstream can be delivered in complete isolation.
- B. Because work such as OT streaming, governance, and gold-layer analytics depends on other platform capabilities being ready first.
- C. Because GitHub does not allow issues without dependencies.
- D. Because ROAM risk management replaces all dependencies.

4. What is the most appropriate role of the Definition of Done in a data-platform program?

- A. To certify that code exists, regardless of validation or documentation.
- B. To ensure that implementation, evidence, quality checks, and delivery documentation meet the agreed completion baseline.
- C. To prevent teams from using GitHub milestones.
- D. To estimate team velocity.

5. In the GitHub mapping described in this article, what is the most appropriate representation of a PI-level planning horizon?

- A. A parent issue.
- B. A pull request.
- C. A milestone.
- D. A commit.

Appendix B. Answer Key

1. **B.** PI Planning aligns teams, dependencies, risks, and commitments around the program mission.
2. **A.** An enabler is technical or architectural work that makes later features viable.
3. **B.** The workstreams are interdependent and must be sequenced visibly.

4. **B.** The DoD ensures work is not only coded, but also validated, documented, and evidenced.
5. **C.** A milestone is the most appropriate representation of a PI-level planning horizon.